

Week 5 — Effect Modification and Interaction

SPPH 604 · Application of Advanced Epidemiological Methods

Running example: obesity, waist fat and death in MASLD (Kueh et al. 2026)

Does waist fat matter more for some people?

The paper's exposure is not one variable but **two crossed**: obesity and waist fat.

Group	Obese?	Waist fat (WHtR)	People	Deaths
I	yes	lower (< 0.6)	355	4 (1.1%)
II	yes	high (≥ 0.6)	4,265	362 (8.5%)
III	no	lower	732	67 (9.2%)
IV	no	high	559	101 (18.1%)

Same 5,911 people as Week 4. WHtR = waist-to-height ratio.

Week 4 asked about **obesity alone**. Today: does **waist fat's** effect depend on obesity — and what do the two do **together**? Two questions that sound alike, and are not.

Our running example

- ▶ **Who:** the Week 4 cohort — US adults with MASLD (NHANES 2007–2018), **5,911** with complete data
- ▶ **Exposure (A): high waist fat**, $\text{WHtR} \geq 0.6$. **Second variable (M): obesity**, $\text{BMI} \geq 30$ (≥ 25 for Asian participants). **Outcome (Y):** death from any cause
- ▶ **L = Week 4's pre-exposure set:** age, sex, race/ethnicity, smoking, sedentary time. Every new model today adjusts for L — not for the paper's Model 1 conditions, some of which Week 4 showed may be mediators
- ▶ **Potential outcomes:** $Y(a)$ when only A is set; $Y(a, m)$ when both are set

The paper compares every group with Group I and never asks either of today's questions. **Today's numbers are our own analysis of the paper's data.**

Read more · Week 4 deck, “Reading the moves with the DAG” · EpiMethods Notation and glossary
· our code: `lectures/analysis/Week5_joint_exposure.R`

By the end of today you can...

1. Tell **effect modification** from **interaction**, and say which one your analysis answers
2. Choose the **adjustment set** each question needs — Week 4's DAGs, one step further
3. Name the **scale** (additive or multiplicative), and explain why the answer can differ
4. **Count** the people and deaths in every cell before trusting an interaction measure
5. Use **time-to-event tools**: a Cox product term, and standardised risks for the additive scale
6. Report a **subgroup finding** without over-reading it

Read more · EpiMethods *Concepts*, “Unpacking Effect Heterogeneity: Interaction vs. Effect Modification”: ehsanx.github.io/EpiMethods/confounding0.html

Part 1 · Two questions that sound alike

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EpiMethods *Concepts*: “Unpacking Effect Heterogeneity” · “Formal Definitions”

Effect modification: one exposure

- ▶ **One exposure, A.** A second variable, M, only sorts people into groups. Is A's effect the **same in every group**?
- ▶ Within $M = m$, compare the two potential outcomes: $RD_m = P[Y(1) = 1 \mid M = m] - P[Y(0) = 1 \mid M = m]$, and RR_m the same as a ratio. Effect modification on the additive scale if $RD_1 \neq RD_0$; multiplicative if $RR_1 \neq RR_0$
- ▶ **MASLD:** does high waist fat's effect on death **differ between the obese and the not obese**?
We imagine changing waist fat only; obesity just sorts people

M's **own** effect needs no causal story: sex can modify an effect though nobody can set it. But M must be something **A cannot change** (Part 2 shows why).

Read more · EpiMethods *Concepts*, “Formal Definitions” · video · VanderWeele, *Epidemiology* 2009 · Karim, *Effect modification in non-randomized studies*, ch. 8, §8.4

Interaction: two exposures, one joint effect

- ▶ **Two exposures, A and M.** Each person now has **four** potential outcomes: $Y(0, 0)$, $Y(1, 0)$, $Y(0, 1)$, $Y(1, 1)$
- ▶ Is the effect of **both together** different from what each does alone — more (synergy) or less (antagonism)?
- ▶ **MASLD:** what would happen to deaths if we could set **both** obesity and waist fat? All four groups are compared with **one** reference

Same 2×2 table, different question. Effect modification imagines changing **one** variable; interaction imagines changing **two**.

Read more · EpiMethods *Concepts*, “Formal Definitions” · Bours, *J Clin Epidemiol* 2021 · Karim, ch. 8, §8.5

Confounding, modification, interaction

	What it is	What you do with it
Confounding	a common cause that biases one estimate	remove it — adjust
Effect modification	A's effect differs across levels of M	report it — one estimate per level
Interaction	A and M together differ from A and M apart	report all four combinations
A product term ($A \times M$)	a term in a model	a tool for both questions, on that model's scale only

One product-term model can serve either question **if it has that question's adjustment set**. What differs is the set (next) and how you **read** the output.

Part 2 · What to adjust for

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EpiMethods *Concepts*: “Implications for Confounding Control”

Each question has its own adjustment set

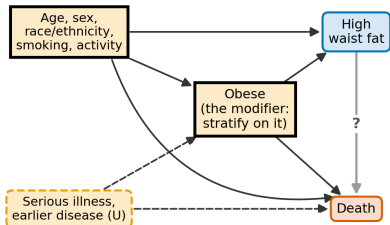
- ▶ **Effect modification of A by M:** adjust for the confounders of $A \rightarrow Y$. M needs none
- ▶ **Interaction of A and M:** adjust for the confounders of $A \rightarrow Y$ and of $M \rightarrow Y$ — a bigger demand
- ▶ Week 4's rule, one step further: **the adjustment set follows the question**

Same two variables, same data: one question can need far more than the other. In MASLD, that decides which question we can even attempt.

Read more · EpiMethods *Concepts*, “Implications for Confounding Control” · effect-modification slides (smoking, income and education) · Karim, ch. 8, Fig. 8.1

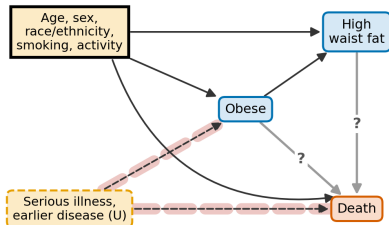
In MASLD, one question needs less

A • Effect modification Does waist fat's effect differ by obesity?



- ✓ {L, obesity} — IF this DAG is right (next slide).
U reaches waist fat only through obesity,
so stratifying on obesity blocks it.

B • Interaction What do obesity and waist fat do together?



- ✗ No valid set, even on this DAG. U confounds
obesity → death (red), and nobody measured it.

Square = conditioned on (adjusted, stratified, restricted or selected). Dashed outline = not measured. Grey arrow with ? = the effect we want.

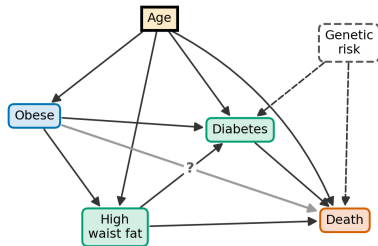
What Panel A's ✓ assumes

- ▶ **Obesity comes before waist fat** (Week 4's Table 2 DAG). If waist fat drives BMI instead, the exposure changes obesity, so obesity cannot be the modifier (two slides on)
- ▶ **U reaches waist fat only by moving people across BMI 30**. Adjusting for BMI itself barely moves today's estimates: 2.80 and 1.07, against 2.99 and 1.09
- ▶ **Nothing unmeasured causes both obesity and waist fat** (a fat-distribution gene, say): obesity is a collider of U and that cause, so stratifying on it would open a path
- ▶ **L is complete**. Week 4's L also had income, missing for 516 of the 5,911

Under all of this sits the **FLI $\geq$ 60 selection**: FLI is computed from BMI **and waist**, so the cohort is selected on a child of both exposures (Week 4). That biases even Panel A — for IV vs III, downward. Interaction fails **before** any of these.

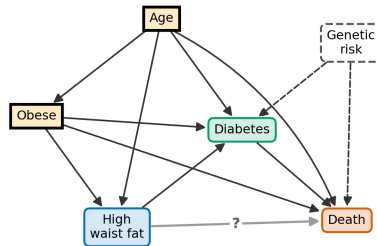
Which way round? Choosing the modifier

Q1 · Effect of obesity



Adjust: age. Not waist or diabetes — both are on obesity's path

Q2 · Effect of high waist fat



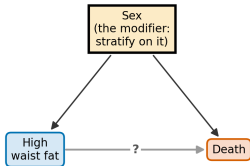
Adjust: age AND obesity (a confounder here). Not diabetes

Square = conditioned on (adjusted, stratified, restricted or selected). Dashed outline = not measured. Grey arrow with ? = the effect we want.

Waist fat by obesity: obesity is a confounder (Q2), so stratifying on it helps. **Obesity by waist fat:** waist fat is on obesity's path (Q1), so stratifying on it removes part of obesity's effect — and this Week 4 figure leaves out U, which also blocks obesity's own effect. Which variable comes first decides which reading works: the DAG, not the data.

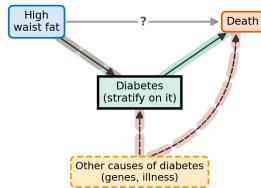
The exposure must not change the modifier

A · Fine: the exposure cannot change it



Waist fat cannot change anyone's sex: each stratum holds the same people, whatever their waist size.

B · Not fine: the exposure can change it



Stratifying on diabetes (1) blocks part of waist fat's effect (green) and (2) opens waist fat $\rightarrow$ diabetes $\leftarrow$ other causes $\rightarrow$ death (red).

Square = stratified on. Dashed outline = not measured. Grey ? = the effect we want. Green = part of the effect; red = an open non-causal path.

“Waist fat's effect among people with $M = m$ ” makes sense only if waist fat cannot move people into or out of that group. So the modifier must be something **the exposure cannot change**. The safe check is time — sex, age, genes, a condition that began **before** the exposure. *Recorded at the same visit* is not enough: waist fat could have caused the diabetes found at the NHANES exam. Our modifier, obesity, passes only if it came before the waist fat.

In R: could waist fat cause the modifier?

```
tab <- table(diabetes = d$t2dm, high_waist = d$central)
tab                                     # people
#>      high_waist
#> diabetes    0    1
#>      0  947 3482
#>      1  140 1342
round(100 * prop.table(tab, 2))      # % in each waist group
#>      high_waist
#> diabetes    0    1
#>      0  87 72
#>      1  13 28
```

Diabetes is twice as common with high waist fat (28% vs 13%). That does not prove waist fat causes it, but it is plausible — so diabetes fails the rule and is a risky modifier.

d: Week 4's 5,911 people, one row each. t2dm: 1 = diabetes at the exam; central: 1 = high waist fat. The "In R" slides are one R session, run in order: `lectures/analysis/Week5_code_outputs.R`.

In R: what splitting by diabetes does

```
f_no <- coxph(Surv(time_yr, dead) ~ central + obese + age + female +
              race + smk_current + sed, data = subset(d, t2dm == 0))
f_yes <- coxph(Surv(time_yr, dead) ~ central + obese + age + female +
              race + smk_current + sed, data = subset(d, t2dm == 1))
round(summary(f_no)$conf.int["central", c(1, 3, 4)], 2) # HR, lower, upper
#>      1.23      0.85      1.78
round(summary(f_yes)$conf.int["central", c(1, 3, 4)], 2)
#>      0.93      0.56      1.54
```

Without diabetes the HR is **1.23 (0.85–1.78)**; with it, **0.93 (0.56–1.54)** — a gap well within noise. And **neither** is waist fat's effect in a fixed group: waist fat moves people into one stratum and out of the other, so both cut the diabetes route and may compare unlike people (panel B).

time_yr: years followed; dead: 1 = died; obese: 1 = obese. L = age, female, race, smk_current, sed (sedentary time, standardised). c(1, 3, 4) picks the HR and its interval.

Part 3 · Which scale?

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EpiMethods *Concepts*: “The Role of the Scale: Effect Measure Modification”

Last week's table, read for modification

Week 4's made-up collapsibility table: half of each age group exposed.

Age group (people)	Risk, exposed	Risk, unexposed	OR	RR	RD
60 or older (200)	0.80	0.50	4.00	1.60	0.30
Under 60 (200)	0.50	0.20	4.00	2.50	0.30

- ▶ Does age **modify** the exposure's effect? **RR: yes** (1.60 vs 2.50). **OR: no** (4.00 in both). **RD: no** (0.30 in both)
- ▶ Same data, three answers. That is why the careful name is **effect-measure modification**

Never write "there is no effect modification" without naming the **scale**.

Read more · EpiMethods *Concepts*, "The Role of the Scale" · Week 4 deck, "Collapsibility: ORs move without confounding"

In R: the same table, three scales

```
toy <- data.frame(age = c("60+", "<60"),
                  risk1 = c(0.80, 0.50), risk0 = c(0.50, 0.20))
toy$RD <- toy$risk1 - toy$risk0           # risk difference
toy$RR <- toy$risk1 / toy$risk0           # risk ratio
toy$OR <- (toy$risk1 / (1 - toy$risk1)) /
          (toy$risk0 / (1 - toy$risk0))   # odds ratio

toy
#>   age risk1 risk0  RD  RR OR
#> 1 60+   0.8   0.5 0.3 1.6  4
#> 2 <60   0.5   0.2 0.3 2.5  4
```

One line per measure. The **RR** differs by age (1.6 vs 2.5); the **OR** (4) and the **RD** (0.3) do not. Same data, three answers — so name the scale.

Two scales, two questions

R_{am} = risk with $A = a$ and $M = m$ (11 = both, 00 = neither); $RR_{am} = R_{am} \div R_{00}$.

Multiplicative — do the **ratios** differ?

- ▶ **ratio of HRs** = HR for A among $M = 1 \div$ HR for A among $M = 0$
- ▶ in a Cox or logistic model: exp(product-term coefficient)
- ▶ **1** = none

Additive — do the **differences** differ?

- ▶ $RD_{M=1} - RD_{M=0}$; for two exposures, the **interaction contrast** $IC = R_{11} - R_{10} - R_{01} + R_{00}$
- ▶ **RERI** (relative excess risk due to interaction) = $RR_{11} - RR_{10} - RR_{01} + 1$
- ▶ **0** = none

- ▶ Two more: **AP** (attributable proportion) = $RERI \div RR_{11}$, 0 = none; **S** (synergy index) = $(RR_{11} - 1) \div [(RR_{10} - 1) + (RR_{01} - 1)]$, 1 = none. RERI, AP and S are all **additive-scale** measures
- ▶ The additive scale counts **deaths you could prevent, and in whom**

Read more · VanderWeele & Knol, “A tutorial on interaction”, *Epidemiol Methods* 2014 ·

Interaction tutorial (oral cancer, alcohol and smoking: OR, RR and RD versions)

Same data, two answers

Risk of the outcome	M absent	M present
A absent	5%	15%
A present	10%	30%

- ▶ **Multiplicative:** RR 2 for A alone, 3 for M alone, 6 for both = $2 \times 3 \rightarrow$ **no interaction**
- ▶ **Additive:** $RERI = 6 - 2 - 3 + 1 = 2$; $IC = 30 - 10 - 15 + 5 = +10$ points $\rightarrow$ **positive interaction**

When both exposures **raise** risk, no multiplicative interaction **means** positive additive interaction:
 $RERI = (RR_{10} - 1)(RR_{01} - 1) > 0$. Neither scale is “right”: **report both**, and say which scale each verdict refers to.

In R: RERI, AP, S and IC from four risks

```
r00 <- 0.05; r10 <- 0.10; r01 <- 0.15; r11 <- 0.30 # risk, A = a, M = m
rr10 <- r10 / r00; rr01 <- r01 / r00; rr11 <- r11 / r00
RERI <- rr11 - rr10 - rr01 + 1
round(c(ratio = rr11 / (rr10 * rr01),          # multiplicative: 1 = none
      RERI = RERI,                             # additive: 0 = none
      AP = RERI / rr11,                       # additive: 0 = none
      S = (rr11 - 1) / ((rr10 - 1) + (rr01 - 1)), # additive: 1 = none
      IC = r11 - r10 - r01 + r00), 3)         # additive, in risk: 0 = none
#> ratio RERI AP S IC
#> 1.000 2.000 0.333 1.667 0.100
```

ratio = 1: no multiplicative interaction. **RERI = 2, AP = 0.33, S = 1.67, IC = +0.10** (ten points): positive additive interaction. Same four risks, two verdicts.

What to report: one table, both scales

Report	Effect modification	Interaction
People and outcomes in every cell	yes	yes
Every combination vs one reference (the lowest-risk cell)	yes	yes
A's effect within each level of M	yes	yes
M's effect within each level of A	—	yes
Measures on both scales, with intervals	yes	yes
What was adjusted for	A's confounders	both exposures' confounders

Read more · Knol & VanderWeele, *Int J Epidemiol* 2012 · EpiMethods *Concepts*, “Reporting guideline” · Karim, ch. 8, §8.7

Part 4 · The four groups, analysed

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EpiMethods: “Interaction” — a joint-exposure analysis on NHANES

Count first

	Lower waist fat	High waist fat
Obese	Group I: 355 people, 4 deaths	Group II: 4,265 people, 362 deaths
Not obese	Group III: 732 people, 67 deaths	Group IV: 559 people, 101 deaths

- ▶ Every comparison **with Group I** divides by **4 deaths** — and Group I is the **lowest-risk** cell, the reference the additive measures need
- ▶ **Week 4's positivity check, for the new exposure:** of 2,432 obese women, only **57** have lower waist fat; among the obese, the chance of high waist fat given L runs from 27%–99.7%

The cell the method needs most is the one the data fill least. **Print these counts** in every interaction table.

5,911 rows, as Week 4.

In R: count first

```
table(group = d$group, died = d$dead)
#>      died
#> group    0    1
#>  I      351    4
#>  II     3903   362
#>  III     665    67
#>  IV      458   101
with(subset(d, obese == 1 & female == 1),      # obese women only
     table(high_waist = central))
#> high_waist
#>    0    1
#>  57 2375
```

Two lines of code, before any model: **4** deaths in Group I, and only **57** of 2,432 obese women with lower waist fat.

5,911 rows. dead: 1 = died during follow-up. central: 1 = high waist fat.

One model, two ways to write it

```
fit_joint <- coxph(Surv(time_yr, dead) ~ group +
                  age + female + race + smk_current + sed, data = d)
fit_prod  <- coxph(Surv(time_yr, dead) ~ obese * central +
                  age + female + race + smk_current + sed, data = d)
all.equal(fit_joint$loglik, fit_prod$loglik)      # the same fit?
#> [1] TRUE
round(exp(coef(fit_prod))[c("obese", "central", "obese:central")], 2)
#>           obese           central obese:central
#>           0.34            1.09            2.74
```

- ▶ **The same fit** (TRUE): the 4-group factor and the product term are two ways of writing one model. Both adjust for $L = \text{age, sex, race/ethnicity, smoking, sedentary time}$
- ▶ obese (0.34) is obesity's HR among those with **lower** waist fat; central (1.09) is waist fat's HR among the **not obese** only; obese:central (2.74) is the **ratio** of waist fat's HR in the obese to that in the not obese
- ▶ Reading central as “the effect of waist fat” is the **Table 2 fallacy** again

Read more · EpiMethods “Interaction” — joint variable and product-term models · Westreich & Greenland, *Am J Epidemiol* 2013 (“...confounder and modifier coefficients”)

In R: which hazard ratio is which

```
b <- coef(fit_prod)
round(exp(b[["central"]]), 2)                # not obese: IV vs III
#> [1] 1.09
round(exp(b[["central"]] + b[["obese:central"]]), 2)  # obese: II vs I
#> [1] 2.99
d$notobese <- 1 - d$obese                    # flip: the obese become the reference
fit_flip <- coxph(Surv(time_yr, dead) ~ notobese * central +
                  age + female + race + smk_current + sed, data = d)
round(summary(fit_flip)$conf.int["central", c(1, 3, 4)], 2)  # HR, lower, upper
#> exp(coef) lower .95 upper .95
#>      2.99      1.11      8.09
```

Add coefficients to get each stratum's HR. For its confidence interval, **flip the coding** so the other stratum is the reference: then `central` is waist fat's HR among the obese, **2.99 (1.11–8.09)**.

In R: does the product term matter?

```
# fit_prod: from "One model, two ways"
fit_main <- coxph(Surv(time_yr, dead) ~ obese + central +      # no product
                  age + female + race + smk_current + sed, data = d)
round(exp(coef(fit_main))[["central"]], 2)    # one HR for all: Week 4's
#> [1] 1.24
lr <- anova(fit_main, fit_prod)               # likelihood-ratio test, product term
round(c(chisq = lr$Chisq[2], p = lr$`Pr(>|Chi|)`[2]), 3)
#> chisq      p
#> 4.754 0.029
```

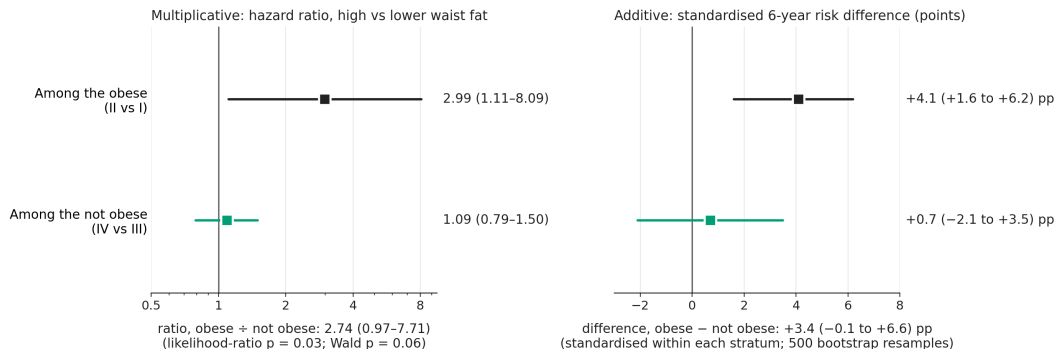
Without the product term, waist fat gets **one** HR for everyone — Week 4's **1.24**. Letting it differ by obesity improves the fit: likelihood-ratio **p = 0.03**. Borderline, and resting on Group I's 4 deaths.

In R: standardised six-year risks

```
risk6 <- function(people, g) { # fit_joint: from "One model, two ways"
  people$group <- factor(g, levels = levels(d$group)) # set everyone to g
  1 - mean(summary(survfit(fit_joint, newdata = people), times = 6)$surv)
}
ob <- subset(d, obese == 1); notob <- subset(d, obese == 0)
r_within <- c(I = risk6(ob, "I"), II = risk6(ob, "II"), # obese
             III = risk6(notob, "III"), IV = risk6(notob, "IV")) # not obese
round(100 * r_within, 1) # standardised 6-year risk, %
#>   I  II III  IV
#> 2.3 6.4 9.1 9.8
rd <- 100 * c(obese = r_within[["II"]] - r_within[["I"]],
             not_obese = r_within[["IV"]] - r_within[["III"]])
round(c(rd, difference = rd[["obese"]] - rd[["not_obese"]]), 1) # points
#>      obese  not_obese difference
#>       4.1       0.7       3.4
```

Week 4's g-computation with a survival curve: set everyone in a stratum to each waist level, predict six-year survival, average. Waist fat adds **4.1** points among the obese and **0.7** among the not obese: a difference of **3.4** points.

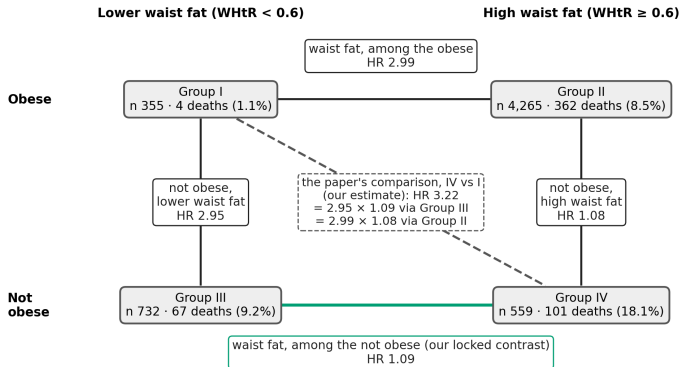
Effect modification: waist fat, by obesity



Week 4's single answer for waist fat, **1.24 (0.93–1.67)**, assumed **one** HR for everyone. Let it differ by obesity: **2.99** among the obese, **1.09** among the not obese. Both scales point the same way. The evidence is **borderline** — likelihood-ratio $p = 0.03$, Wald interval 0.97–7.71 — and the obese estimate rests on Group I's 4 deaths.

5,911 rows, adjusted for L. Risk differences are standardised six-year risks (Week 4's g-computation): within each stratum, everyone set to lower, then high, waist fat.

The paper's IV vs I, taken apart



Lines are comparisons, not causes. HRs adjusted for L; 5,911 rows.

- ▶ IV vs I changes **both** exposures, and splits two ways: 2.95×1.09 via Group III, or 2.99×1.08 via Group II. Which step looks big depends on the route — because the two **interact**, today's topic
- ▶ Either way, the big step is the one that **leaves Group I** and its 4 deaths. The one-exposure-at-a-time contrast is our locked **IV vs III: 1.09**

3.22 is our estimate of the paper's comparison, adjusted for L. The paper reports 2.9 (its Model 2).

Interaction: four cells, one reference

*HR (95% CI) vs Group I, the lowest-risk cell, adjusted for L · deaths in the cell. The two “exposures” are now **not being obese** (A) and **high waist fat** (M): RR_{10} = Group III, RR_{01} = Group II, RR_{11} = Group IV.*

	Lower waist fat	High waist fat	Waist fat, within row
Obese	1 (reference) · 4 deaths	2.99 (1.11–8.09) · 362	2.99 (1.11–8.09)
Not obese	2.95 (1.07–8.13) · 67	3.22 (1.17–8.87) · 101	1.09 (0.79–1.50)
Not obese, within column	2.95 (1.07–8.13)	1.08 (0.86–1.35)	

- ▶ **Multiplicative:** $HR_{IV} \div (HR_{II} \times HR_{III}) = \mathbf{0.37 (0.13–1.03)}$ — the forest plot's 2.74, coded the other way ($0.37 = 1 \div 2.74$, up to rounding)
- ▶ **Additive:** $RERI = \mathbf{-1.72 (-4.57 \text{ to } 1.13)}$. Standardised 6-year risks with everyone in Group I, II, III or IV: **2.5%, 7.0%, 6.9%, 7.5%** → $IC = \mathbf{-3.9 (-6.7 \text{ to } -0.5)}$ points

The point estimates are **less than additive** — the same borderline evidence as before, coded the other way. Every comparison with Group I leans on its 4 deaths (1.09 and 1.08 do not). And this **cannot be read as causal**: U, obesity's confounder, was never measured.

In R: RERI, AP, S and IC for the four groups

```
# fit_joint and risk6(): from the earlier In-R slides
h <- exp(coef(fit_joint))[c("groupII", "groupIII", "groupIV")] # vs I
RERI <- h[["groupIV"]] - h[["groupII"]] - h[["groupIII"]] + 1
round(c(RERI = RERI, AP = RERI / h[["groupIV"]],
        S = (h[["groupIV"]] - 1) /
            ((h[["groupII"]] - 1) + (h[["groupIII"]] - 1))), 2)

#>  RERI    AP    S
#> -1.72 -0.53 0.56

r_all <- sapply(c("I", "II", "III", "IV"),
               function(g) risk6(d, g)) # everyone, set to each group
round(100 * c(r_all, IC = r_all[["IV"]] - r_all[["II"]] -
              r_all[["III"]] + r_all[["I"]]), 1)

#>    I    II   III   IV   IC
#>  2.5  7.0  6.9  7.5 -3.9
```

Read the four risks in pairs. If everyone were obese, high waist fat would add $7.0 - 2.5 = 4.5$ points; if no one were, $7.5 - 6.9 = 0.6$. $IC = 0.6 - 4.5 = -3.9$: waist fat adds more among the obese — the standardised-risk slide's pattern, now averaged over everyone. Both point estimates are below additive; RERI's interval includes 0 (-4.57 to 1.13), IC's does not (-6.7 to -0.5).

Part 5 · Tools for time-to-event, and a subgroup claim

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EpiMethods *Concepts*: “Reporting guideline”

The right tools for a time-to-event outcome

Scale	Tool	Effect modification (waist fat, obese vs not)	Interaction (Group I reference)
Multiplicative	product term in the Cox model → ratio of HRs	2.74 (0.97–7.71)	0.37 (0.13–1.03)
Additive	standardised risks at a fixed time → difference of RDs, or IC	+3.4 (−0.1 to +6.6) points	IC −3.9 (−6.7 to −0.5) points
Additive, shortcut	RERI from HRs — close to the risk-based RERI when deaths are uncommon	—	−1.72 (risk-based −1.56)

- ▶ **Don't** turn survival into “ever died” and fit a logistic model: it throws away follow-up time and censoring
- ▶ A **binary** outcome with fixed follow-up (Thursday's lab) is different: logistic is right there. But an OR approximates an RR only when the outcome is **rare**, so with a common outcome an RERI from ORs is **not** the RERI from risks

A dramatic subgroup claim

Crude HR, Group IV vs Group I, fitted separately in men and women (the 6,048-row locked domain, where Group I has 375 people):

	Men	Women
HR, IV vs I	36.14 (8.86–147.48)	4.11 (0.99–17.12)
Group I, deaths / people	2 / 317 (0.6%)	2 / 58 (3.4%)

- ▶ “Non-obese waist fat is far deadlier in men”? Each HR **divides by a death rate built on 2 deaths**
- ▶ The crude ratio, women vs men, is 0.11 (0.02–0.85): “significant” — for a contrast that changes **both** exposures, in a crude model

Count first. A “significant” interaction can rest on four deaths.

In R: count Group I by sex first

```
with(subset(dom, group == "I"), table(female, died = dead)) # 6,048 rows
#>      died
#> female  0   1
#>      0 315   2
#>      1  56   2
```

Two deaths among the men in Group I, and **two** among the women. Every IV-vs-I hazard ratio in the last table divides by one of these death rates.

dom: the 6,048-row locked domain, before rows with incomplete covariates are dropped — the rows P3's code uses.

The same question, asked properly

Our locked contrast, IV vs III, with sex as the modifier: one test of the product term.

Model	Ratio of HRs, women $\div$ men	p
Crude (P3; 1,296 rows)	0.62 (0.26–1.43)	0.278
The paper's Model 1 (P3; 1,296 rows)	1.00 (0.42–2.35)	0.993
Week 4's pre-exposure set (1,291 rows)	1.13 (0.48–2.66)	0.78

- Additive (P3): six-year risk differences of +0.8 (−3.3 to +4.2) points in men and +0.5 (−7.3 to +7.6) in women — small point estimates, wide intervals

No evidence of modification at this precision — and an interval compatible with the effect in women being **less than half**, or **more than twice**, that in men. Report the interval; don't explain a gap the data cannot.

p: likelihood-ratio tests. 1,296 = Groups III + IV complete on Model 1; 1,291 = Groups III + IV within Week 4's 5,911.

In R: one test for sex, on the locked contrast

```
s34 <- subset(d, group %in% c("III", "IV")) # the locked contrast only
s34$IV <- as.integer(s34$group == "IV")
f0 <- coxph(Surv(time_yr, dead) ~ IV + female +          # no product term
            age + race + smk_current + sed, data = s34)
f1 <- coxph(Surv(time_yr, dead) ~ IV * female +          # IV x sex
            age + race + smk_current + sed, data = s34)
round(summary(f1)$conf.int["IV:female", c(1, 3, 4)], 2) # HR, lower, upper
#> exp(coef) lower .95 upper .95
#>      1.13      0.48      2.66
round(anova(f0, f1)$`Pr(>|Chi|)`[2], 2)                # likelihood-ratio p
#> [1] 0.78
```

Restricting to Groups III and IV makes the product term test exactly the locked contrast: **1.13 (0.48–2.66)**, likelihood-ratio **p = 0.78**. No evidence of modification, and an interval wide enough to allow large differences either way.

Honest subgroup reporting

1. **Pre-specify** one modifier, with a reason. One well-argued modifier beats twenty
2. Say **which question** — effect modification or interaction. It sets the adjustment set
3. **Count** the people and outcomes in every cell
4. Report **both scales**, with intervals, and name the reference cell
5. A big crude gap is not evidence of modification; a null test is not evidence of none
6. Don't **explain** a difference the data cannot

Read more · Knol & VanderWeele, *Int J Epidemiol* 2012 · Karim, ch. 8, §8.7–8.8 · Interaction tutorial: effect modification

Key takeaways

1. **Effect modification:** one exposure, its effect differs by M — and M must be something the exposure **cannot change**. **Interaction:** the joint effect of two exposures
2. Modification needs A's confounders; interaction needs **both** exposures'. In MASLD, interaction fails even under the assumptions that would rescue modification
3. **Name the scale.** Harmful exposures with no multiplicative interaction have additive interaction
4. **Count every cell.** The lowest-risk reference is often the sparse one
5. Time-to-event: a **Cox product term** and **standardised risks**, not “ever died”
6. Waist fat may matter more among the obese — borderline, on four deaths. The paper's IV vs I mixes two exposures and cannot be split uniquely

Next: Week 6 — survey weights. Design-aware, P3's Model 1 sex ratio (1.00) becomes 2.53 (0.92–6.98) (M1). **Exit question:** one modifier for your own paper — and why your exposure cannot change it.

Where to read more

Topic	EpiMethods and course pages	Key reading
Definitions	<i>Concepts</i> — Formal Definitions · video · slides	VanderWeele 2009; Bours 2021
What to adjust for	<i>Concepts</i> — Implications for Confounding Control	Karim, ch. 8
Scale	<i>Concepts</i> — The Role of the Scale	VanderWeele & Knol 2014
Joint exposure, NHANES	Interaction (joint variable, product term, RERI)	Knol et al. 2011
Worked OR / RR / RD Reporting	Interaction tutorial <i>Concepts</i> — Reporting guideline	Hosmer & Lemeshow 1992 Knol & VanderWeele 2012
Thursday's lab	Exercise · solution	—

References

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- ▶ Lajous M, Bijon A, Fagherazzi G, et al. Body mass index, diabetes, and mortality in French women: explaining away a “paradox”. *Epidemiology* 2014;25:10–14.
- ▶ Knol MJ, VanderWeele TJ. Recommendations for presenting analyses of effect modification and interaction. *Int J Epidemiol* 2012;41:514–520.
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- ▶ Kueh MTW, et al. Body weight categories and fat distribution in relation to all-cause mortality among adults with metabolic dysfunction-associated steatotic liver disease. *BMJ Open* 2026;16:e113719.
- ▶ VanderWeele TJ. On the distinction between interaction and effect modification. *Epidemiology* 2009;20:863–871.
- ▶ VanderWeele TJ, Knol MJ. A tutorial on interaction. *Epidemiol Methods* 2014;3:33–72.
- ▶ Westreich D, Greenland S. The Table 2 fallacy: presenting and interpreting confounder and modifier coefficients. *Am J Epidemiol* 2013;177:200–209.

Appendix: the two questions as dagitty code

Paste into the *Model code* box at dagitty.net. With Waist as the only exposure, dagitty returns {**L**, **Obese**}. Mark Obese as an exposure too, and it reports that **no adjustment set exists** — until you remove the latent tag from U.

```
dag {  
  Waist [exposure]  Death [outcome]  U [latent]  
  L -> Obese  L -> Waist  L -> Death  
  U -> Obese  U -> Death  
  Obese -> Waist  Obese -> Death  Waist -> Death  
}
```

L = age, sex, race/ethnicity, smoking, sedentary time. U = serious illness and disease that came before obesity (Week 4's Ill and Before). Then test the assumptions: add G [latent] G -> Obese G -> Waist, or FLI [adjusted] Waist -> FLI Waist -> TG TG -> FLI TG -> Death, and watch every adjustment set disappear.